普通物理学学习笔记

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目录

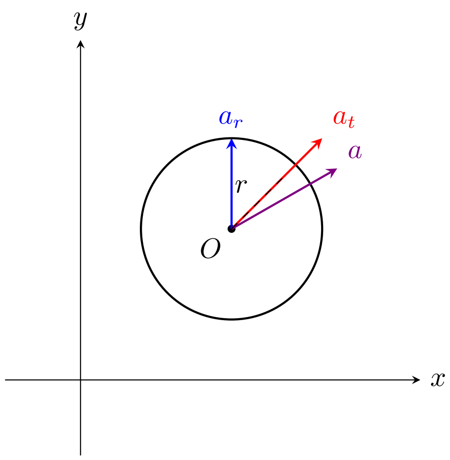
1 切向加速度与法向加速度

1.1 切向（tangential）与法向（radial）加速度

$$a_t = \frac{dv}{dt} = r\frac{d\omega}{dt} = r\alpha \tag{1}$$

$$a_r = \frac{v^2}{r} = r\omega^2 \quad \text{(向心加速度)} \tag{2}$$

$$a = r\sqrt{\alpha^2 + \omega^4} \tag{3}$$



2 刚体转动的功与功率

$$dW = \vec{F} \cdot d\vec{s} = rd\theta \cdot F \sin\varphi = \tau d\theta \tag{4}$$

瞬时功率（**Instantaneous power**）：

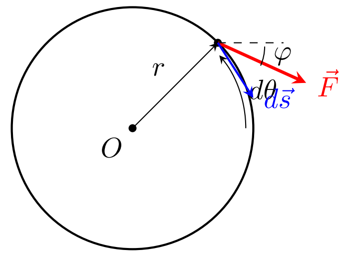
$$P = \frac{dW}{dt} = \tau \frac{d\theta}{dt} = \tau\omega \tag{5}$$

（类比 $P = Fv$）

$$\sum \tau = I\alpha = I\frac{d\omega}{dt} = I\frac{d\omega}{d\theta}\frac{d\theta}{dt} = I\omega\frac{d\omega}{d\theta} \tag{6}$$

$$\Rightarrow \sum \tau d\theta = I\omega d\omega \tag{7}$$

$$\Rightarrow dW = I\omega d\omega \quad \Rightarrow \quad W = \frac{1}{2}I\omega^2 \tag{8}$$

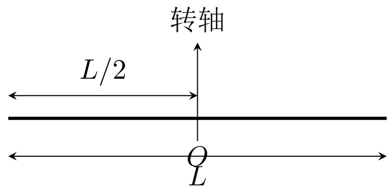


3 常见刚体的转动惯量

3.1 细杆绕中心（Rod about center）

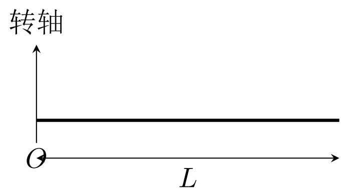
$$I = \int r^2 dm = \int_{-L/2}^{L/2} r^2 \cdot \frac{M}{L} dr = \frac{1}{12}ML^2 \tag{9}$$

（$dm = \frac{M}{L}dr$）



3.2 细杆绕一端（Rod about one end）

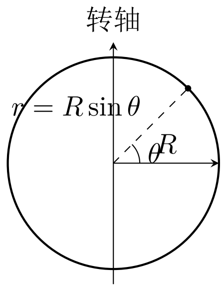
$$I = \int r^2 dm = \int_0^L r^2 \cdot \frac{M}{L} dr = \frac{1}{3}ML^2 \tag{10}$$



3.3 球壳（Spherical shell）

$$I = \int r^2 dm = \int_0^{\pi} (R\sin\theta)^2 \frac{M\sin\theta d\theta}{2} = \frac{2}{3}MR^2 \tag{11}$$

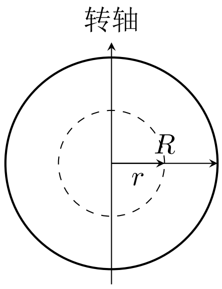
（$\dfrac{dm}{M} = \dfrac{2\pi R\sin\theta \cdot Rd\theta}{4\pi R^2} = \dfrac{\sin\theta d\theta}{2}$）



3.4 实心球（Solid sphere）

$$I = \int \frac{2}{3}r^2 dm = \int_0^{R} \frac{2}{3}r^2 \cdot \frac{3Mr^2}{R^3}dr = \frac{2}{5}MR^2 \tag{12}$$

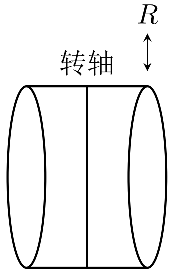
（$dm = \rho \cdot 4\pi r^2 dr = \dfrac{3Mr^2}{R^3}dr$，球壳 $dI = \frac{2}{3}r^2 dm$）



3.5 实心圆柱（Solid cylinder）

$$I = \int r^2 dm = \int_0^{R} r^2 \cdot \frac{2Mr}{R^2}dr = \frac{1}{2}MR^2 \tag{13}$$

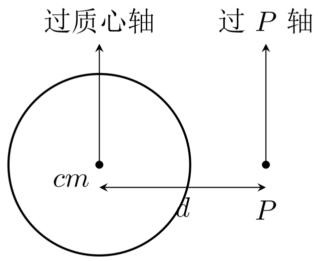
（$dm = \dfrac{M}{\pi R^2 h} \cdot 2\pi rhdr = \dfrac{2Mr}{R^2}dr$）



4 平行轴定理（Parallel-Axis Theorem）

$$I_P = I_{cm} + Md^2 \tag{14}$$

其中 d 为两平行轴间的距离。



5 一些刚体模型

5.1 旋转杆（Rotating rod）

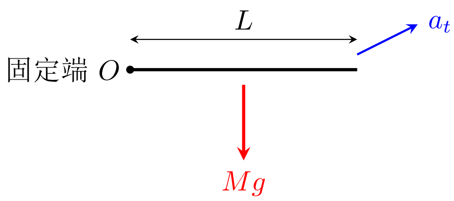
长度 L，质量 M，均匀分布，绕一端转动。

由转动定律：

$$\tau = \frac{L}{2} \cdot Mg = I\alpha = \frac{1}{3}ML^2\alpha \tag{15}$$

$$\Rightarrow \alpha = \frac{3g}{2L} \tag{16}$$

$$a_t = L\alpha = \frac{3}{2}g \tag{17}$$



6 滑轮转动（考虑滑轮半径 R、转动惯量 I，求加速度 a）

方法一（转动定律）：

$$(T_1 - T_2)R = I\alpha = I\frac{a}{R} \tag{18}$$

$$m_1 g - T_1 = m_1 a, \quad T_2 - m_2 g = m_2 a \tag{19}$$

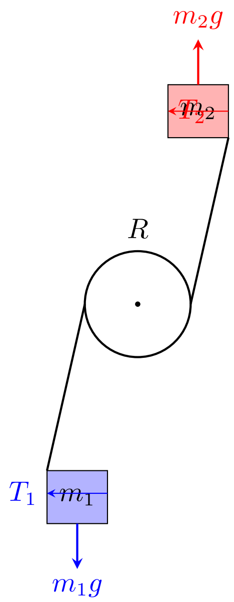
联立可解 T_1, T_2, a。

方法二（能量守恒）：

$$\Delta K + \Delta U = 0 \tag{20}$$

$$\Delta K = \frac{1}{2}(m_1 + m_2)v^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}\left(m_1 + m_2 + \frac{I}{R^2}\right)v^2 \tag{21}$$

$$\Delta U = (m_2 - m_1)gh \tag{22}$$



7 矢量式与参考系

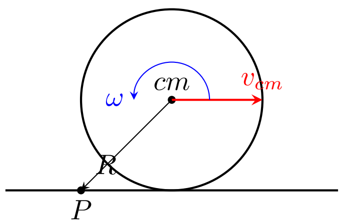
$$\vec{v}=\vec{\omega}\times\vec{r},\quad \vec{\tau}=\vec{r}\times\vec{F},\quad \vec{L}=\vec{r}\times\vec{p} \tag{23}$$

科里奥利力（Coriolis Force）：$-2m(\vec{\omega}\times\vec{v}_r)$

纯滚动（无滑动）：

$$v_{cm}=\omega R \tag{24}$$

接触点 P 的瞬时速度为零。



8 角动量的应用

$$\sum\vec{\tau}_{ext}=\frac{d\vec{L}}{dt} \tag{25}$$

若 $\sum\vec{\tau}_{ext}=0$，则角动量守恒 $\vec{L}=\text{const}$。

$$\vec{L}=\sum\vec{r}_i\times(m_i\vec{v}_i),\quad L_z=I\omega \tag{26}$$

$$\frac{dL_z}{dt}=I\alpha=\sum\tau_{ext,z} \tag{27}$$

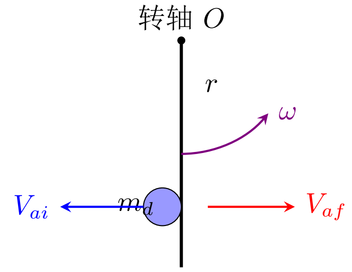
（注意一般 $\vec{L}\neq I\vec{\omega}$，仅当绕对称轴转动时成立。）

例：子弹打杆（子弹质量 m_d，杆质量 m_s，绕端点转动惯量 I）：

$$m_d V_{ai} = m_d V_{af} + m_s V_s \tag{28}$$

$$-r m_d V_{ai} = -r m_d V_{af} - I\omega \tag{29}$$

$$\frac{1}{2} m_d V_{ai}^2 = \frac{1}{2} m_d V_{af}^2 + \frac{1}{2} m_s V_s^2 + \frac{1}{2} I\omega^2 \tag{30}$$

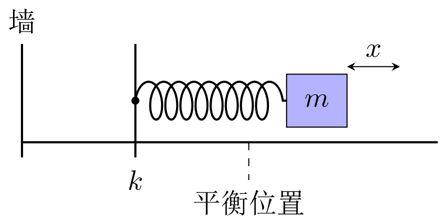


9 简谐振动（SHM）

$$m\frac{d^2x}{dt^2} = -kx \quad \Rightarrow \quad \frac{d^2x}{dt^2} = -\omega^2 x, \quad \omega = \sqrt{\frac{k}{m}} \tag{31}$$

$$x = A\cos(\omega t + \varphi) \tag{32}$$

$$T = \frac{2\pi}{\omega}, \quad f = \frac{\omega}{2\pi}, \quad E = \frac{1}{2}kA^2 \tag{33}$$



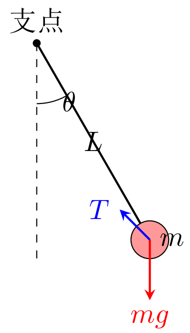
10 振动模型

10.1 单摆（Simple Pendulum）

$$-mg\sin\theta = mL\frac{d^2\theta}{dt^2} \tag{34}$$

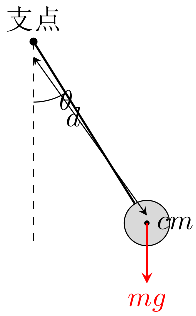
小角度时 $\sin\theta \approx \theta$：

$$\omega = \sqrt{\frac{g}{L}}, \quad \theta = \theta_0 \cos(\omega t + \varphi) \tag{35}$$



10.2　物理摆（Physical Pendulum）

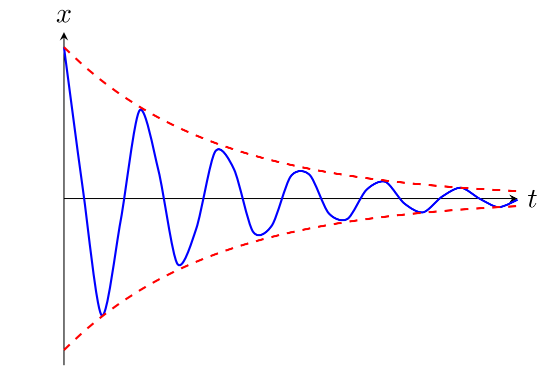
$$\omega = \sqrt{\frac{mgd}{I}} \tag{36}$$



10.3　阻尼振动（Damped Oscillator）

$$m\ddot{x} + b\dot{x} + kx = 0 \tag{37}$$

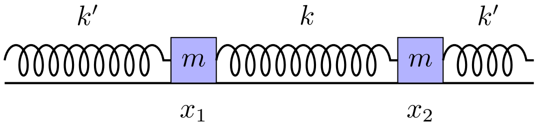
$$x = Ae^{-\frac{b}{2m}t}\cos(\omega t + \varphi), \quad \omega = \sqrt{\frac{k}{m} - \left(\frac{b}{2m}\right)^2} \tag{38}$$



10.4　简正模式（Normal Modes）

两质量 m 由弹簧 k 相连，两端弹簧 k'：

$$\omega = \sqrt{\frac{k'+k}{m}}\ (\text{同相}), \quad \omega = \sqrt{\frac{k'}{m}}\ (\text{反相}) \tag{39}$$



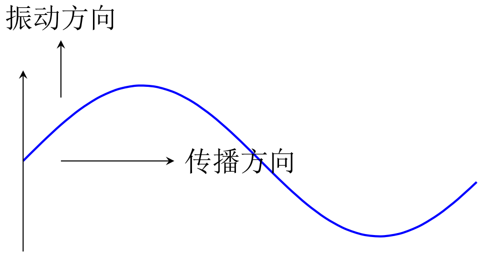
10.5　受迫振动与共振（Forced Oscillation）

$$m\ddot{x} + b\dot{x} + kx = F_{ext}\cos(\omega t) \tag{40}$$

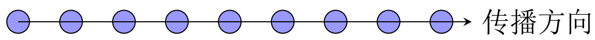
共振时 $\omega = \omega_0 = \sqrt{k/m}$，振幅最大，相位差 $\pi/2$。

11　波

11.1　横波与纵波



横波（振动 $\perp$ 传播）



纵波（振动 $\parallel$ 传播）

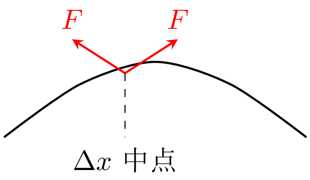
11.2　波动方程与波速

$$\frac{\partial^2 y}{\partial t^2} = v^2 \frac{\partial^2 y}{\partial x^2}, \quad v = \frac{\omega}{k} = \frac{\lambda}{T} \tag{41}$$

11.3　弦上波速

$$v = \sqrt{\frac{F}{\mu}} \tag{42}$$

（F 为张力，μ 为线密度）



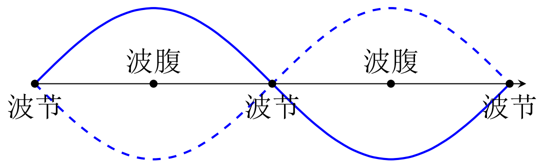
11.4　干涉

$$\text{相长}: \Delta r = n\lambda, \qquad \text{相消}: \Delta r = (n + \tfrac{1}{2})\lambda \tag{43}$$

拍频：$f_b = |f_1 - f_2|$

11.5　驻波（Standing waves）

$$y = 2A \sin(kx) \cos(\omega t) \tag{44}$$



12 声波与多普勒效应

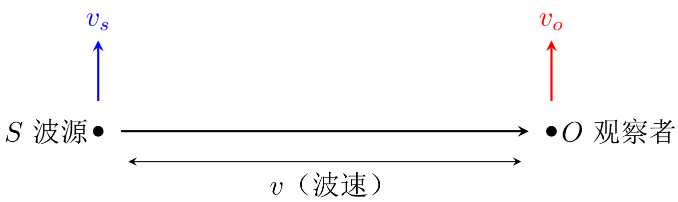
12.1 声速与声强

$$v = \sqrt{\frac{B}{\rho}}, \quad I = \frac{1}{2}B\omega k A^2 \tag{45}$$

12.2 多普勒效应

$$f' = \frac{v + v_o}{v - v_s} f \tag{46}$$

v_o：观察者速度，v_s：波源速度（正方向约定从观察者指向波源）。



13 相对论

13.1 光速不变与速度合成

$$w = \frac{u + v}{1 + \frac{uv}{c^2}} \tag{47}$$

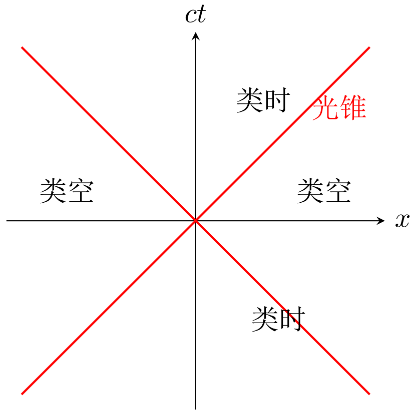
13.2 时间膨胀与长度收缩

$$\Delta t = \gamma \Delta t_0, \quad L = \frac{L_0}{\gamma}, \quad \gamma = \frac{1}{\sqrt{1 - v^2/c^2}} \tag{48}$$

13.3 洛伦兹变换

$$x' = \gamma(x - vt), \quad t' = \gamma(t - \frac{vx}{c^2}), \quad y' = y, \ z' = z \tag{49}$$

13.4 光锥（Minkowski 图）



13.5 质能关系

$$E = mc^2 = \gamma m_0 c^2, \quad E_0 = m_0 c^2, \quad E^2 = p^2c^2 + m_0^2 c^4 \tag{50}$$

对光子：$E = h\nu$，$p = h\nu/c$。

14 热学

14.1 热膨胀

$$\Delta L = \alpha L_0 \Delta T, \quad \Delta V = \beta V_0 \Delta T, \quad \beta \approx 3\alpha \tag{51}$$

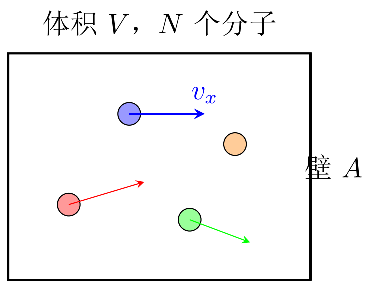
14.2 理想气体状态方程

$$PV = nRT = Nk_BT \tag{52}$$

范德瓦尔斯方程：$(P + \frac{an^2}{V^2})(V - nb) = nRT$

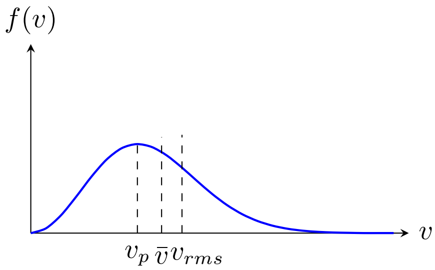
14.3 气体动理论

$$P = \frac{1}{3}\rho\overline{v^2}, \quad \frac{1}{2}m\overline{v^2} = \frac{3}{2}k_BT \tag{53}$$



14.4 麦克斯韦速度分布

$$f(v) = 4\pi \left(\frac{m}{2\pi k_B T} \right)^{3/2} v^2 e^{-\frac{mv^2}{2k_B T}} \tag{54}$$



15 热力学第一定律

$$\Delta U = Q - W \tag{55}$$

15.1 等温、等压、等体过程

$$W_{iso} = nRT \ln \frac{V_f}{V_i}, \quad C_p = C_v + R, \quad \gamma = \frac{C_p}{C_v} \tag{56}$$

15.2 绝热过程

$$PV^\gamma = \text{const}, \quad TV^{\gamma-1} = \text{const} \tag{57}$$

15.3 热机效率与卡诺循环

$$e = \frac{W}{Q_H}, \quad e_{Carnot} = 1 - \frac{T_c}{T_h} \tag{58}$$